

Customer Support Chatbot — Architecture Document

AgentCore Managed Harness Implementation

Overview

This project routes customer support messages using the **AgentCore managed harness** rather than a Bedrock Flow. All classification, information-gathering, and grounding behavior lives in a single *system_prompt.txt* file — there are no Condition nodes or a separate classifier resource. This is consistent with the project's Instructions page: Bedrock Agents Classic, which a Flow's Agent node depends on, closed to new AWS customers on July 30, 2026.

Architecture Diagram

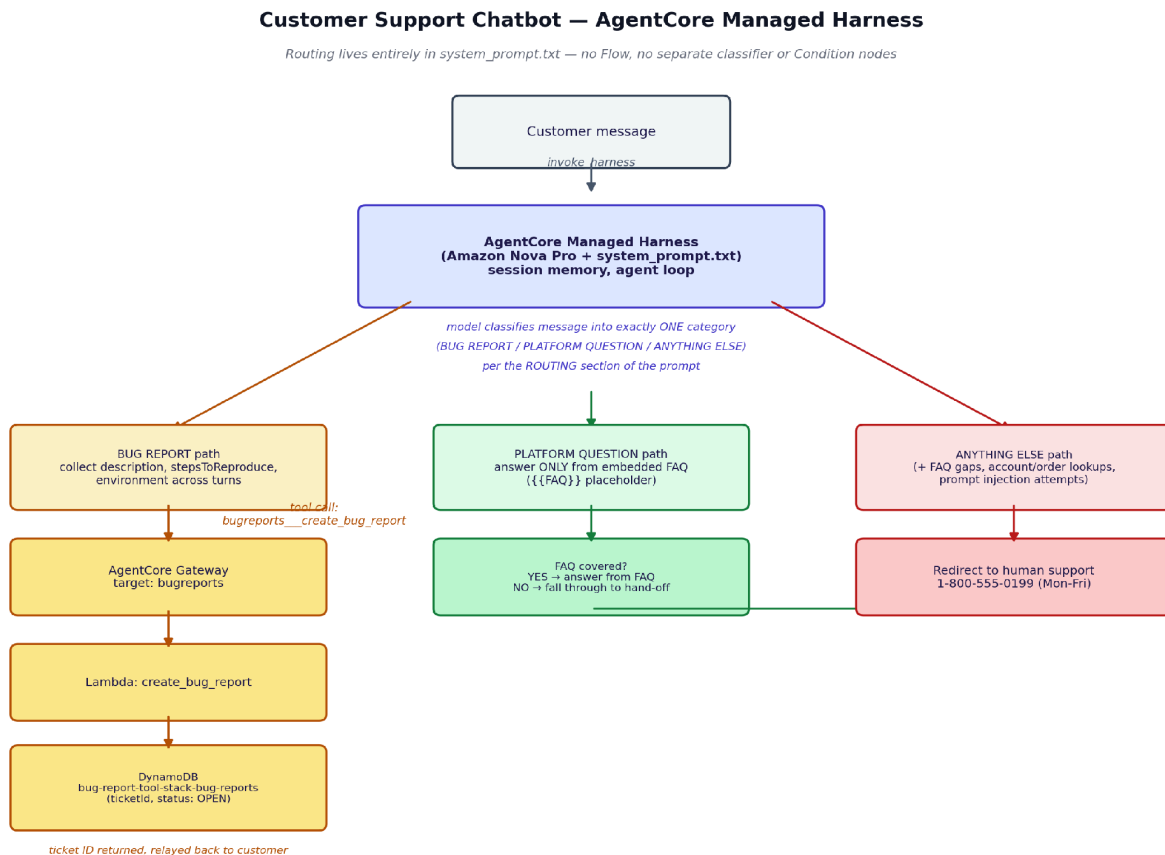


Figure 1. Customer messages flow through the AgentCore harness, showing all three routing paths and the bug-report tool chain (Gateway → Lambda → DynamoDB).

Component Summary

AgentCore Managed Harness

Runs the full agent loop server-side: model calls (Amazon Nova Pro, pinned via `us.amazon.nova-pro-v1:0`), session memory, and tool execution. Created and updated via `create_harness.py`, which also substitutes the `{{FAQ}}` placeholder in `system_prompt.txt` with the contents of `online_shop_faq.md` at build time.

Routing (inside `system_prompt.txt`)

Every incoming message is classified by the model into exactly one of three categories before any response is generated:

- BUG REPORT — collects description, steps to reproduce, and environment across turns before calling the tool.
- PLATFORM QUESTION — answered only from the embedded FAQ; falls through to hand-off if uncovered.
- ANYTHING ELSE — includes out-of-scope requests, account/order-specific lookups, and prompt-injection attempts; redirects to human support.

Bug Report Tool Chain

The harness invokes `create_bug_report` through an AgentCore Gateway target named "bugreports" (visible in transcripts as the tool call `bugreports__create_bug_report`). The Lambda persists the ticket to the `bug-report-tool-stack-bug-reports` DynamoDB table and returns a `ticketId`, which is relayed back to the customer.

Why No Flow / Condition Nodes

The project's original rubric describes a Bedrock Flow with a classifier node and Condition-node routing. That design requires a classic Bedrock Agent for the bug-report path, which cannot be created on AWS accounts opened after July 30, 2026. The Instructions page for this project specifies the AgentCore harness as the replacement architecture; this document and the accompanying evidence reflect that architecture.